AI Resume Reviewer & Feedback Platform

Comprehensive Project Report & Review Guide | Automated ATS Analysis & Optimization

Project Domain: Artificial Intelligence / NLP / Full-Stack Web	**Version:** 1.1.0 (Production Ready)
Target Audience: Job Seekers, Students, University Career Cells	**Review Phase:** First Project Review & Architecture Evaluation

1. Executive Summary & Problem Statement

The Problem: Over 75% of job applicant resumes are automatically rejected by corporate Applicant Tracking Systems (ATS) before any human recruiter reads them. Common causes include unparsed document formatting, absent core skill keywords, lack of quantifiable impact metrics in bullet points, and mismatch against target job descriptions. Furthermore, job seekers receive generic rejection notices without actionable diagnostic feedback.

The Solution: The *AI Resume Reviewer & Career Feedback Platform* is a specialized intelligent web platform designed to function as an automated virtual career mentor. It extracts, audits, scores, and rewrites resumes against 15 industry-standard technology career tracks in real time, empowering candidates to identify critical skill gaps and optimize their resume prior to actual company submissions.

2. Core Features & Capabilities

- **Explainable ATS Scoring Engine (0-100):** Generates a composite score based on skill match (40%), target role alignment (30%), resume section completeness (15%), and keyword density (15%).
- **Dual Role Analysis Modes:** Supports 'With Target Role' (evaluates against a chosen career path like Frontend Developer or Data Scientist) and 'Without Target Role' (auto-detects the candidate's optimal tech track using Machine Learning).
- **Rule-Based NLP Skill Extraction:** Audits resumes against 200+ industry technical skills and classifies detected skills versus missing required skills for the role.
- **Target Job Description Matcher:** Computes TF-IDF Cosine Similarity between a candidate's resume and any raw job posting text, providing exact match percentages and keyword gap analysis.
- **Interactive Real-Time Text Editor:** Allows users to live-edit resume text with real-time recalculation of scores and instant feedback without requiring file re-uploads.
- **AI Bullet Point Rewriter:** Transforms passive duty descriptions ('worked on website') into high-impact, quantified XYZ achievement statements ('Engineered responsive UI using React, driving an 18% lift in conversion').
- **ATS Layout & Format Verification:** Validates page count compliance (1-page standard), contact details existence (email, phone, LinkedIn/GitHub), and standard section headings.
- **Supabase Cloud Database Persistence:** Stores every review session, extracted text, feedback, and score into Supabase PostgreSQL, enabling historical retrieval and progress tracking.

3. Complete Technology Stack & Specifications

Layer	Technology / Framework	Functional Role & Justification
Frontend	React 18, Vite 5, Tailwind CSS 3	High-performance client UI with Hot Module Replacement, modular component architecture, and responsive styling.
Icons & UI	Lucide React, Axios	Modern vector iconography and reliable asynchronous HTTP communication with FastAPI backend.
Backend API	FastAPI (Python 3.10+), Uvicorn	Asynchronous ASGI web framework offering sub-millisecond route execution and automated OpenAPI documentation.
Validation & I/O	Pydantic v2, Python-Multipart, HTTPX	Strict data schemas, high-speed multipart PDF file uploads, and asynchronous cloud requests.
PDF Processing	PyMuPDF (fitz)	High-speed C-native PDF text extraction, layout reading, and font/metadata analysis without Java dependencies.
Machine Learning	Scikit-Learn (LinearSVC)	Support Vector Classification trained on tech resumes for 15 career track domain predictions with low latency.
NLP & Vectorization	TF-IDF Vectorizer, Regex	Text tokenization, n-gram weighting, cosine similarity matching, and deterministic skill pattern extraction.
Database	Supabase (PostgreSQL)	Cloud PostgreSQL instance with Row Level Security (RLS) policies for feedback and resume storage.
Deployment	Vercel (Frontend), Render (Backend)	Global CDN edge delivery for React SPA and dedicated containerized Python runtime with Dockerfile/Procfile.

4. End-to-End System Workflow

1. **Document Intake:** Candidate uploads a PDF resume. The system validates file signature, size (max 10MB), and format.
2. **Text & Structure Parsing:** PyMuPDF extracts raw text, page count, and font layout metrics while Regex checks for contact links.
3. **Feature Extraction & Classification:** Text is normalized and vectorized using TF-IDF. LinearSVC predicts career domain (or validates candidate target role).
4. **Competency Audit:** The rule-based NLP engine scans for technical skills, computing intersection (matched) and set difference (missing).
5. **Score Computation:** The composite scoring formula aggregates skill relevance, section presence, and keyword frequency into a 0-100 rating.
6. **Persistence & Display:** Results are committed to Supabase PostgreSQL, and the frontend dynamically renders score gauges, feedback cards, and rewrites.

5. First Review Q&A; Guide (Expected Questions & Answers)

Q1: Why did you choose LinearSVC over deep learning models like BERT or LSTM?

Answer: LinearSVC with TF-IDF features is optimized for sparse, high-dimensional text classification. It trains in seconds, runs inference in under 5 milliseconds, uses less than 2 MB of RAM, and requires zero GPU infrastructure. For classifying resumes into defined career tracks, it delivers state-of-the-art accuracy without the latency and deployment complexity of large language models.

Q2: How does the system handle resumes with missing or varied section titles?

Answer: The parser implements flexible regular expressions that accommodate industry synonyms (e.g., 'Work Experience', 'Employment History', 'Professional Background'). If a key section is absent, the section score is penalized, and an actionable recommendation is surfaced to the candidate.

Q3: How is the Job Description Matcher calculated?

Answer: Both the resume text and the job description are transformed into n-gram TF-IDF vector spaces. We compute the Cosine Similarity between the two vectors, which quantifies semantic and vocabulary alignment on a normalized 0% to 100% scale.

Q4: How does data persistence work?

Answer: Feedback, scores, detected skills, and extracted text are securely saved into a Supabase PostgreSQL database using PostgREST endpoints. The client frontend connects via a 'Database History' drawer to retrieve and reload previous reviews instantly.

> ***Project Philosophy:*** *"A resume is more than a document; it is a story of what someone is capable of doing. Through our AI, we transform those capabilities into something tangible for every candidate striving toward their dream career."*